Ex. No. 09	ACCESSING APIs THROUGH HTTP
Date of Exercise	
Git Hub Host Link	https://github.com/gurubruhh/Flutter-Laboratory.git

Aim

To access a REST API using HTTP and display JSON data in Flutter.

Description

This experiment demonstrates fetching data from an external API using Flutter. It uses HTTP GET requests to retrieve JSON data and decode it into readable information. The fetched data is displayed using ListView.builder(). Loading indicators and error messages are also implemented.

Procedure:

1. Check Flutter installation using flutter doctor.
2. Create a Flutter project using flutter create api_app.
3. Open the project using VS Code.
4. Add the HTTP package in pubspec.yaml.
5. Import dart:convert and the HTTP package.
6. Create an API function using an HTTP GET request.
7. Fetch data from the JSONPlaceholder API.
8. Decode the JSON response using jsonDecode().
9. Display the fetched data using ListView.builder() with loading and error handling.
10. Run the application and verify the API output.

Program

```
import 'package:flutter/material.dart';
import 'dart:convert';
import 'package:http/http.dart' as http;

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
```

```
      home: const ApiScreen(),
    );
  }
}

class ApiScreen extends StatefulWidget {
  const ApiScreen({super.key});

  @override
  State<ApiScreen> createState() => _ApiScreenState();
}

class _ApiScreenState extends State<ApiScreen> {
  List data = [];
  bool isLoading = true;
  String errorMessage = '';

  @override
  void initState() {
    super.initState();
    fetchData();
  }

  // Fetch API Data
  Future<void> fetchData() async {
    try {
      final response = await http.get(
        Uri.parse('https://jsonplaceholder.typicode.com/posts'),
      );

      if (response.statusCode == 200) {
        setState(() {
          data = jsonDecode(response.body);
          isLoading = false;
        });
      } else {
        setState(() {
          errorMessage = 'Failed to load data';
          isLoading = false;
        });
      }
    } catch (e) {
      setState(() {
        errorMessage = 'Error: Check your internet connection';
        isLoading = false;
      });
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
```

```
      appBar: AppBar(
        title: const Text('API Data Fetch'),
        backgroundColor: Colors.blue,
        foregroundColor: Colors.white,
      ),

      body: isLoading
          ? const Center(child: CircularProgressIndicator())
          : errorMessage.isNotEmpty
              ? Center(child: Text(errorMessage))
              : ListView.builder(
                  itemCount: data.length,
                  itemBuilder: (context, index) {
                    return Card(
                      margin: const EdgeInsets.all(8),
                      child: ListTile(
                        leading: CircleAvatar(
                          child: Text('${data[index]['id']}'),
                        ),
                        title: Text(data[index]['title']),
                        subtitle: Text(data[index]['body']),
                      ),
                    );
                  },
                ),
    );
  }
}
```

Output

DartPad
Create
Create with Gemini
Samples
Reload
Run
import 'package:flutter/material.dart';
import 'dart:convert';
import 'package:http/http.dart' as http;
void main() {
runApp(const MyApp());
}
class MyApp extends StatelessWidget {
const MyApp({super.key});
@override
Widget build(BuildContext context) {
return MaterialApp(
debugShowCheckedModeBanner: false,
home: const ApiScreen(),
);
}
}
class ApiScreen extends StatefulWidget {
const ApiScreen({super.key});
@override
State<ApiScreen> createState() => _ApiScreenState();
}
API Data Fetch
sunt aut facere repellat provident occaecati excepturi optio reprehenderit
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suscipit recusandae consequuntur expedita et cum
reprehenderit molestiae ut ut quas totam
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voluptatem doloribus vel accusantium quis pariatur
molestiae porro eius odio et labore et velit aut
3
eum et est occaecati
ullam et saepe reiciendis voluptatem adipisci
sit amet autem assumenda provident rerum culpa
quis hic commodi nesciunt rem tenetur doloremque ipsam iure
quis sunt voluptatem rerum illo velit
4
Ask Gemini to change your code or app!
Privacy notice
Feedback
Dart 3.13.7 • Flutter 3.47.9
Stable channel

Result

Thus, the Flutter application was successfully developed to load and display image and video assets. The video was successfully controlled using Play and Pause buttons.